

Technical Summary - Tic-Tac-Toe Project

1. Project Overview

Tic Tac Toe is a console-based game developed using Python. The project provides two modes: Computer Mode and Two players Mode. It allows users to play traditional 3x3 Tic Tac Toe Game.

The project demonstrate the use of lists, functions, loops, conditional statements, user input, global variables, and the random module.

2. Project Statement

The project aims to develop a program that can manage a complete Tic Tac Toe game by:

- Displaying a 3x3 board
- Accepting and validating player moves
- Preventing occupied positions from being selected
- Switching turns between players.
- Detecting wins and draws.
- Providing a computer opponent.

3. Technology Used

- Programming Language : Python
- Interface : Command Line
- Module : random

4. Data Structure

The board is represented using a list of nine elements:

```
board = [
    "-", "-", "-",
    "-", "-", "-",
    "-", "-", "-"
]
```

Each element represents a position on the board.

'-' indicates an empty position, while X and O represent the players.

The main variables are:

- board - Stores the board state.
- currentplayer - Stores the active player.
- winner - Stores the winning player.
- game running - controls the game loop.

5. Main Functions

The program is divided into functions for modularity:

- printboard() - displays the board.
- playerinput() - accepts and validates moves.
- checkhorizontal() - checks horizontal wins.
- checkvertical() - checks vertical wins.
- checkdiagonal() - checks diagonal wins.
- checkwin() - determines the winner.
- checktie() - detects a draw.
- switchplayers() - changes the current player.
- computer() - generates a random move.
- twoplayermode() - manages two-player gameplay.
- computermode() - manages computer gameplay.

6. Working of the System

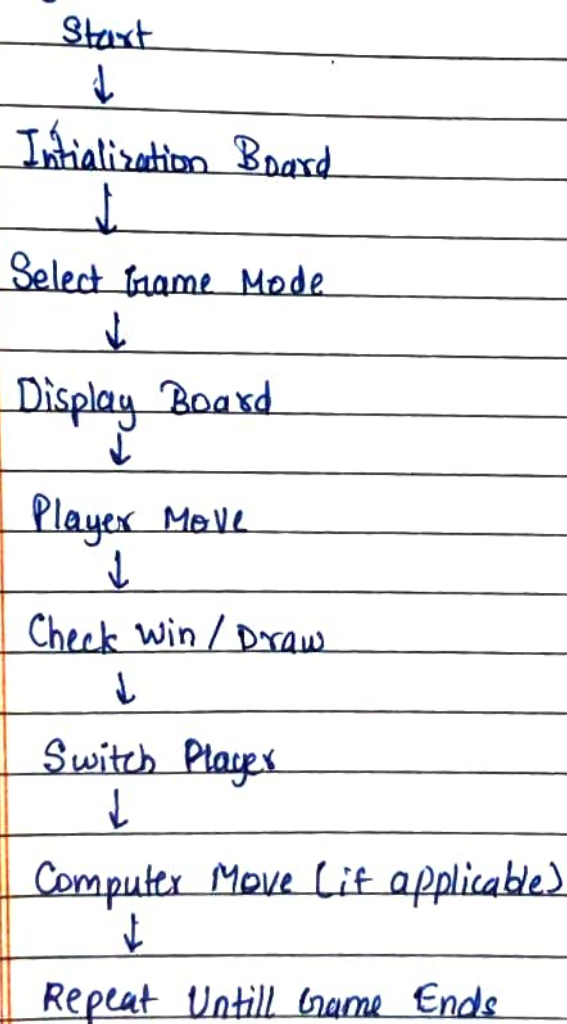
The program first displays a menu allowing the user to select Computer Mode or Two Player Mode.

After selecting a mode, the board is displayed and the player enters a position from 1-9. The input is validated and the board is ~~enter~~ updated if the position is empty.

After every move, the program checks for a win or draw. If neither occurs, the turn is switched.

In Computer Mode, the computer uses: `random.randint(0, 8)` to select an empty position and place O.

Program Flow:



8. Technical Features

~~Technical~~ The program uses modular programming to divide the game into independent functions. A list is used for efficient board representation, while conditional statements identify winning combinations:

Basic validation prevents invalid positions and overwriting of existing moves. The computer opponent uses random selection rather than an AI strategy.